



COVID-19 myths busted!

WHO busts some prevalent myths concerning the novel coronavirus.
<https://digit.in/may20-61>



New COVID-19 symptoms

The CDC has confirmed six new symptoms of the novel coronavirus including loss of smell or taste. <https://digit.in/may20-62>

be treated, there was a shift towards studying how exactly the viruses affected populations.

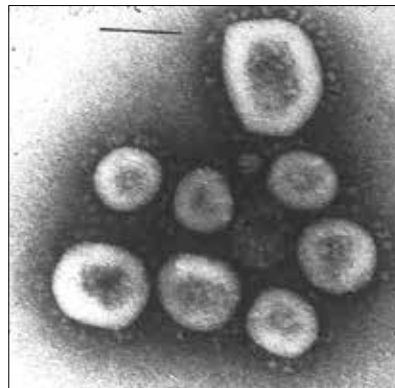
Because viruses and bacteria evolve so rapidly, variations are tracked by researchers who called them serotypes. Research into the various serotypes of coronaviruses showed that some, but not all were more infectious during the winter months than the summer months, in the upper latitudes. In the tropical countries, there is no seasonal variation of incidence. Others displayed periodic outbreaks in populations, with intervals of between two to three years. While some viruses caused highly local infections, others were across large portions of land. Some coronaviruses continuously circulate within the human population, which can be considered as a prolonged outbreak. Most of the coronaviruses that infected humans were shown to originate from bats. The rates for reinfection were also high, and it showed that the same virus could infect the same person even after the immune system had tackled it and flushed it out of the body before. While the infection could occur at any age, most coronaviruses infected children more easily on exposure. The coronaviruses caused respiratory illnesses, pneumonia in infants and young adults, exacerbated asthma in children, and caused bronchitis in the elderly. Coronaviruses were also studied in a wide range of creatures, including rabbits, turkeys and cats.

The coronaviruses are classified into four groups called alphacoronaviruses, betacoronaviruses, gammacoronaviruses and deltacoronaviruses. While the first two affect mostly mammals, including humans, the latter two affect mostly birds. There are some exceptions though, such as gammacoronaviruses that affect whales, and deltacoronaviruses that infect pigs.

All of these findings indicate just how infectious and difficult to tackle the coronaviruses are,

and these are for the viruses that are considered mild. This was the state of the coronavirus research in 2000. It was considered an obtuse undertaking, a career dead end because there was no real progress even after forty years of research. Coronaviruses, anyway, caused only a fraction of the cases of common cold as compared to other viruses, such as the rhinovirus.

In 2002, the Severe acute respiratory syndrome (SARS) epidemic broke out with the first case identified in Guangdong, China. In 2003, the World Health Organisation identified a species of Coronavirus as the cause of the outbreak. The



Coronavirus imaged in 1967

Credit: Proceedings of the National Academy of Sciences, USA

coronavirus that caused SARS was the first of five human coronaviruses to be identified since the SARS outbreak, which revitalised coronavirus research. Although the exact source was never pinpointed, there was a lot of research in interspecies spread of the coronavirus, because of the possibility that the SARS virus could have originated from bats. The same virus was also found in civets, cats, raccoons and badgers. Most of the human coronaviruses start off from bats or rates, then move on to intermediate domestic animal hosts such as cows, pigs or camels, before being transmitted to humans.

Since 2003, five new coronavirus that affect humans have been identified, and many more that affect animals. There have been Middle East respiratory syndrome (MERS)

outbreaks in 2012, 2015 and 2018, and finally the ongoing 2019 SARS-CoV-2 outbreak. Animals also suffer from coronavirus outbreaks, with the bird flu and the swine flu being two well known examples.

There are now seven known strains of human coronaviruses. The two identified before the SARS outbreak were HCoV-229E and HCoV-OC43. SARS-CoV-1 was identified in 2003, followed by HCoV-NL63 in 2004, HCoV-HKU1 in 2005, MERS-CoV in 2012, and finally SARS-CoV-2 in 2019. The HCoV in the names stand for Human Coronavirus. SARS and MERS stand for the respective diseases that they cause. The letters indicate the location in which the virus was first identified, Netherlands and Hong Kong. Out of these SARS-CoV-1, MERS-CoV and SARS-CoV-2 are the coronaviruses that cause symptoms that are considered severe, and are the more dangerous variants of the coronavirus. The SARS outbreak saw about 8000 people infected, with a mortality rate of 10 percent. All the MERS outbreaks together resulted in 2,468 cases, and a mortality rate of 34 percent. The SARS-CoV-2 however, has infected in excess of 2.5 million people, and counting.

Despite the increased interest in research into coronaviruses since the SARS epidemic, very little is known about the overall clinical impact of each of the variations of coronaviruses. It is easy to make individuals, even in large populations, simply sit out seasonal outbreaks. In the higher latitudes, the health care system actually prepares for coronavirus spikes during the winter months. Even more severe outbreaks such as SARS can wash over some regions, without much research or counter-measures taken by local authorities, at least in time to prevent the epidemic. All of this only goes to show that we need to learn more about this family of viruses, and dedicate more resources towards research to investigate them further. **1**